

Predicting Coronary Heart Disease in Adult

Using Support Vector Machines: A Comparative Analysis of Linear, Radial, and Polynomial Kernels

Introduction

Cardiovascular disease is a leading cause of death worldwide, making early prediction essential. This study uses Support Vector Machine (SVM) models to predict coronary heart disease based on demographic and lifestyle factors such as age, BMI, physical activity, sleep, and sex. Using data from the National Health Interview Survey (IPUMS NHIS, 2024), we compare linear, RBF, and polynomial kernels to evaluate how model complexity affects performance. The analysis also considers class imbalance to better assess model reliability

Data Preparation

- Adults (18+)
- Binary target: Heart Disease
- Features: Age, BMI, Activity, Sleep, Sex

References

Methodology

Data was cleaned, standardized, and split into training/testing sets using key health variables. Linear, RBF, and polynomial SVM models were tuned with cross-validation, with class imbalance handled using class weights and performance evaluated using multiple metrics and visualizations.

Feature Importance

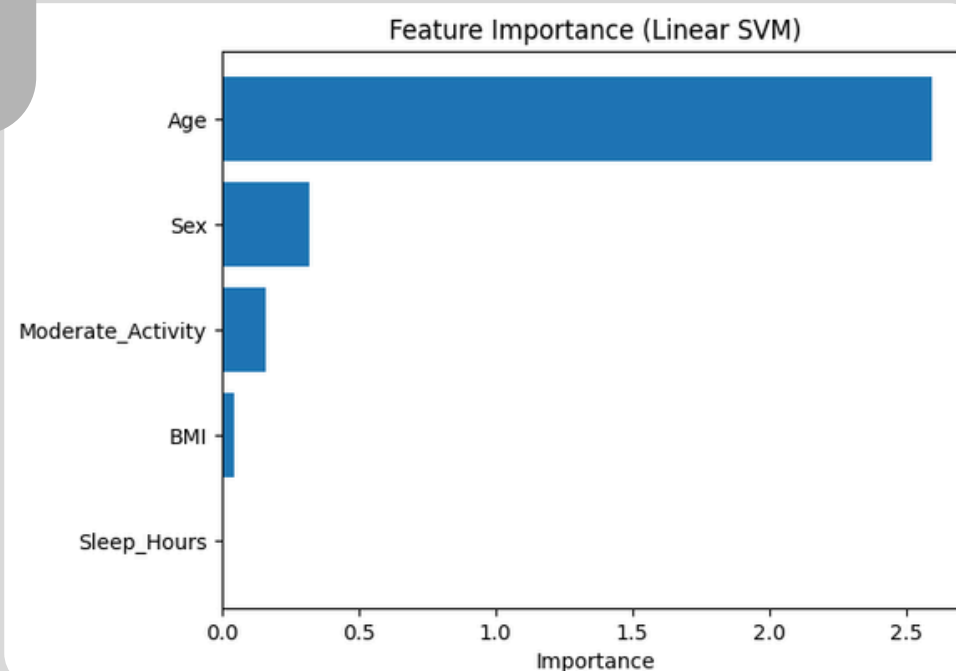


Figure: Feature Importance for Heart Disease Prediction

SVM Algorithm

Support Vector Machines (SVM) find an optimal decision boundary that maximizes the margin between two classes. The linear decision function is: $f(x) = w^T x + b$, where w is the weight vector and b is the bias. The model minimizes: $(1/2)||w||^2 + C \sum \xi_i$ subject to: $y_i(w^T x_i + b) \geq 1 - \xi_i$ where C controls the trade-off between maximizing the margin and minimizing classification error, ξ_i are slack variables that allow for misclassification. Support vectors are the key data points that define the decision boundary, lying on or within the margin.

Result

Three SVM models were trained and tuned, with the polynomial model performing best (accuracy = 0.76, lowest error). Class imbalance affected results—linear and RBF models had high recall but low precision, while the polynomial model achieved better balance. Age was the most important predictor, with overall weak feature separation observed in visualizations.

Age and Heart Disease Relationship

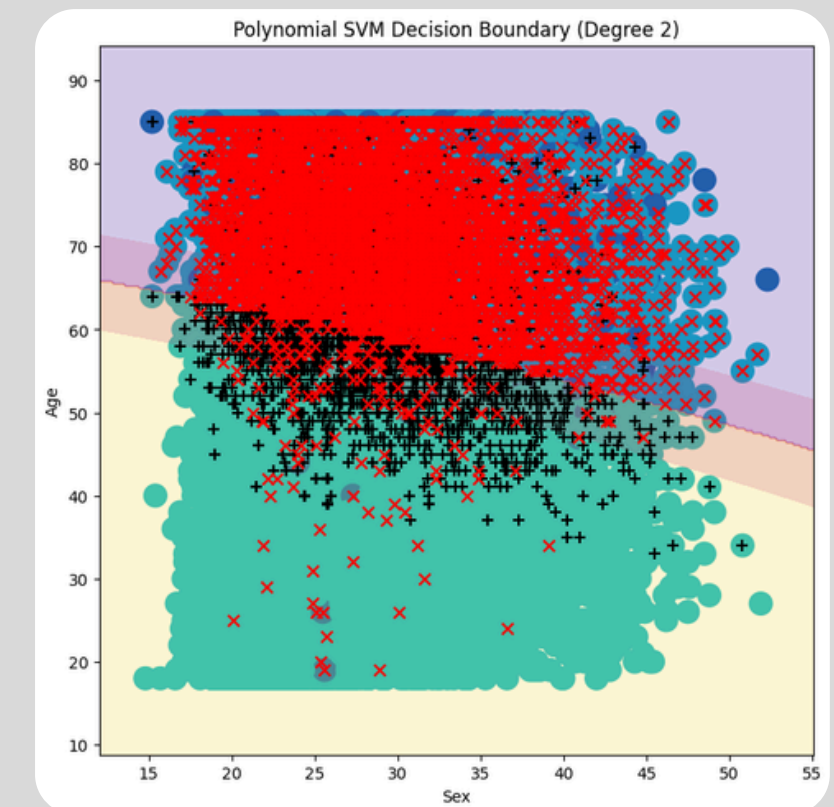


Figure: Age influences risk, overlap indicates multiple factors

Discussion and Conclusion

Polynomial SVM performs best (accuracy 0.76), while linear and RBF show similar results (~0.65). Class imbalance affects performance—linear/RBF have high recall but low precision, while polynomial is more balanced. Weak feature relationships and overlap limit model accuracy; age is the strongest predictor.